

GEOGRAPHY

PHYSICAL GEOGRAPHY



GEOMORPHOLOGY

MODULE - 28

- **GEOMORPHIC PROCESSES
WEATHERING- PHYSICAL, CHEMICAL
AND BIOLOGICAL (PART-2)**

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



Heat is a major weathering force in Death Valley, New Mexico.

TEMPERATURE CHANGES AND EXPANSION

- 1 With rise in temperature, mineral expands and pushes against its neighbor and as temperature falls, a corresponding contraction takes place.
- 2 Because of diurnal changes in the temperatures, this internal movement among the mineral grains of the superficial layers of rocks takes place regularly.
- 3 This process is most effective in dry climates and high elevation where diurnal temperature changes are drastic.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES

TEMPERATURE CHANGES AND EXPANSION



Exfoliation on a rounded boulder

4 The surface layers of the rocks tend to expand more than the rock at depth and this leads to formation of stress within the rock resulting in fracturing parallel to the surface.

5 Due to differential heating and resulting expansion and contraction of surface layers and their subsequent exfoliation from the surface results in smooth rounded surfaces in rocks.

6 In rocks like granites, smooth surfaced and rounded small to big boulders called tors form due to such exfoliation.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES

EXAMPLES OF EXFOLIATION OR ONION WEATHERING PROCESS

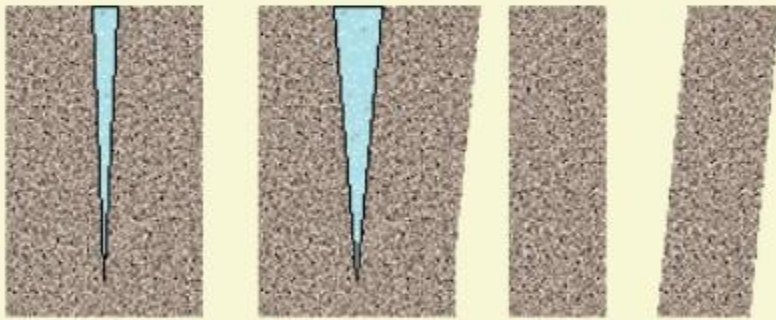


WEATHERING

2

PHYSICAL WEATHERING PROCESSES

Frost Wedging



Water-filled
crack

Freezes to
ice

Breaks
Rock

FREEZING, THAWING AND FROST WEDGING

- 1 Frost weathering occurs due to growth of ice within the pores and cracks of rocks during repeated cycles of freezing and melting.
- 2 This process is most effective at high elevations in mid-latitudes where freezing and melting is often repeated.
- 3 Glacial area are subject to frost wedging daily. In this process, the rate of freezing is important.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



FREEZING, THAWING AND FROST WEDGING

4

Rapid freezing of water causes its sudden expansion and high pressure.

5

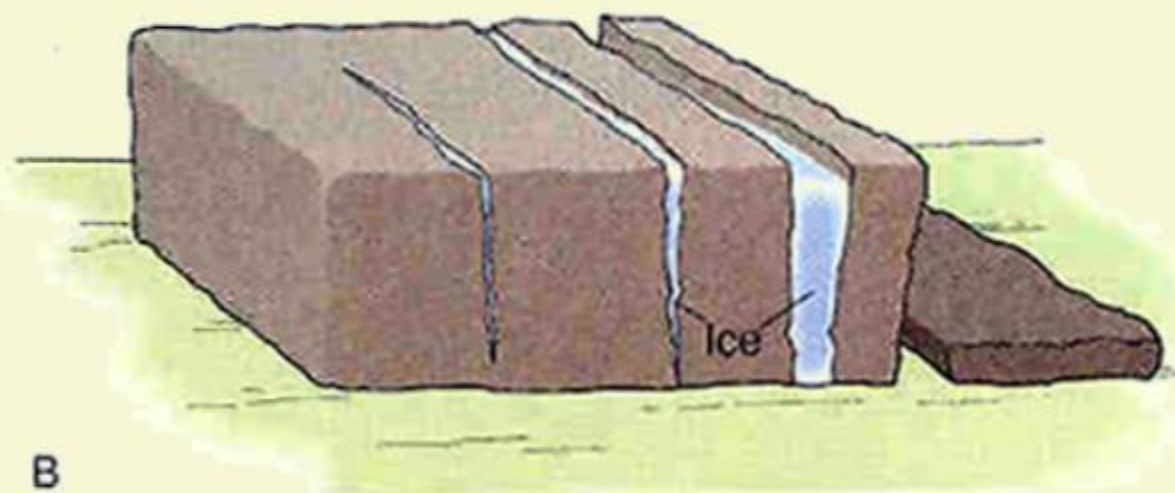
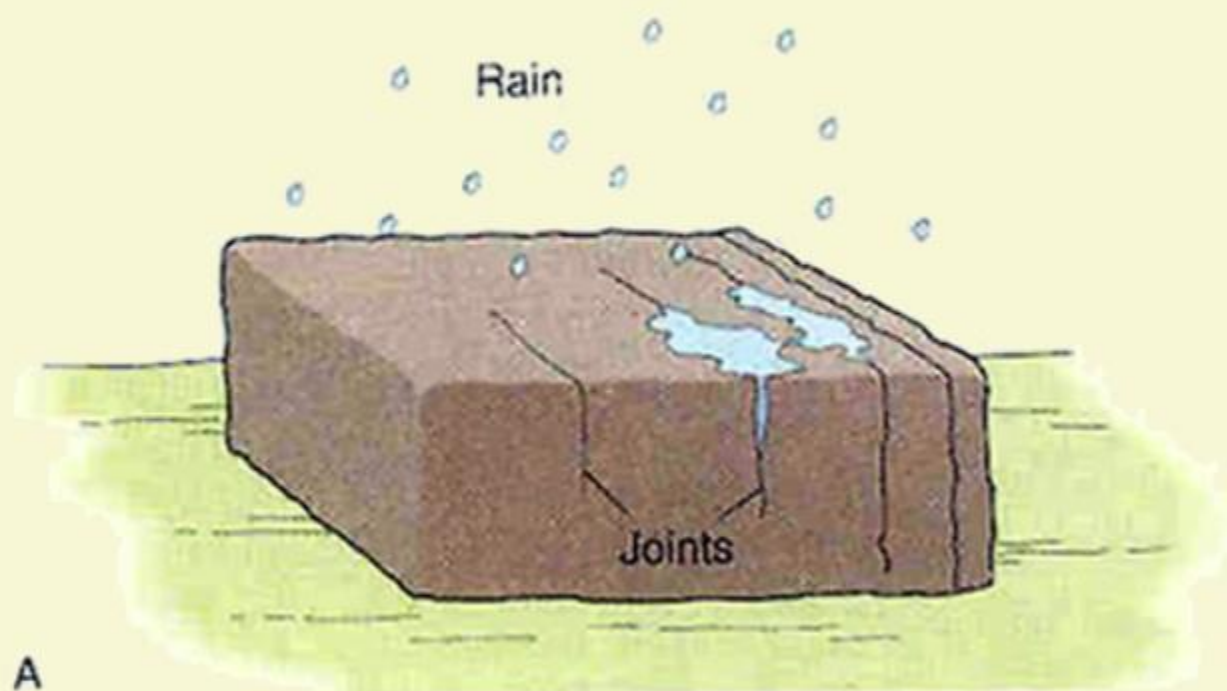
The resulting expansion affects joints, cracks and small inter granular fractures to become wider and wider till the rock breaks apart.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES

FROST WEDGING PROCESS



A
Frost Wedging

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



SALT WEATHERING

- 1 Salt in rocks expand due to thermal action, hydration and crystallization. Many salts like calcium, sodium, magnesium, potassium and barium have a tendency to expand.
- 2 Expansion of these salts depends on temperature and their thermal properties.
- 3 High temperature ranges between 30 and 50⁰ C of surface temperatures in deserts favour such salt expansion.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



SALT WEATHERING

4

Salt crystals in near-surface pores cause **splitting of individual grains** within rocks, which eventually fall off.

5

This process of **falling off of individual grains** may result in **granular disintegration** or granular foliation. Salt crystallisation is the most effective of all salt-weathering processes.

WEATHERING

2

PHYSICAL WEATHERING PROCESSES



Exfoliation on a rounded boulder

SALT WEATHERING

6

Sodium chloride and gypsum crystals in desert areas heave up overlying layers of materials and with the result polygonal cracks develop all over the heaved surface.

7

With salt crystal growth, chalk breaks down most readily, followed by limestone, sandstone, shale, gneiss and granite etc.

WEATHERING

3

BIOLOGICAL WEATHERING PROCESSES



1 Biological weathering is contribution to or removal of minerals from the weathering environment and physical changes due to growth or movement of organisms.

2 Burrowing and wedging by organisms like earthworms, termites, rodents etc. help in exposing the new surfaces to chemical attack and assists in the penetration of moisture and air.

3 Human beings by disturbing vegetation, ploughing and cultivating soils also help in mixing and creating new contacts between air, water and minerals in the Earth materials

WEATHERING

3

BIOLOGICAL WEATHERING PROCESSES



4 Decaying plant and animal matter help in the production of humic, carbonic and other acids which enhance decay and solubility of some elements.

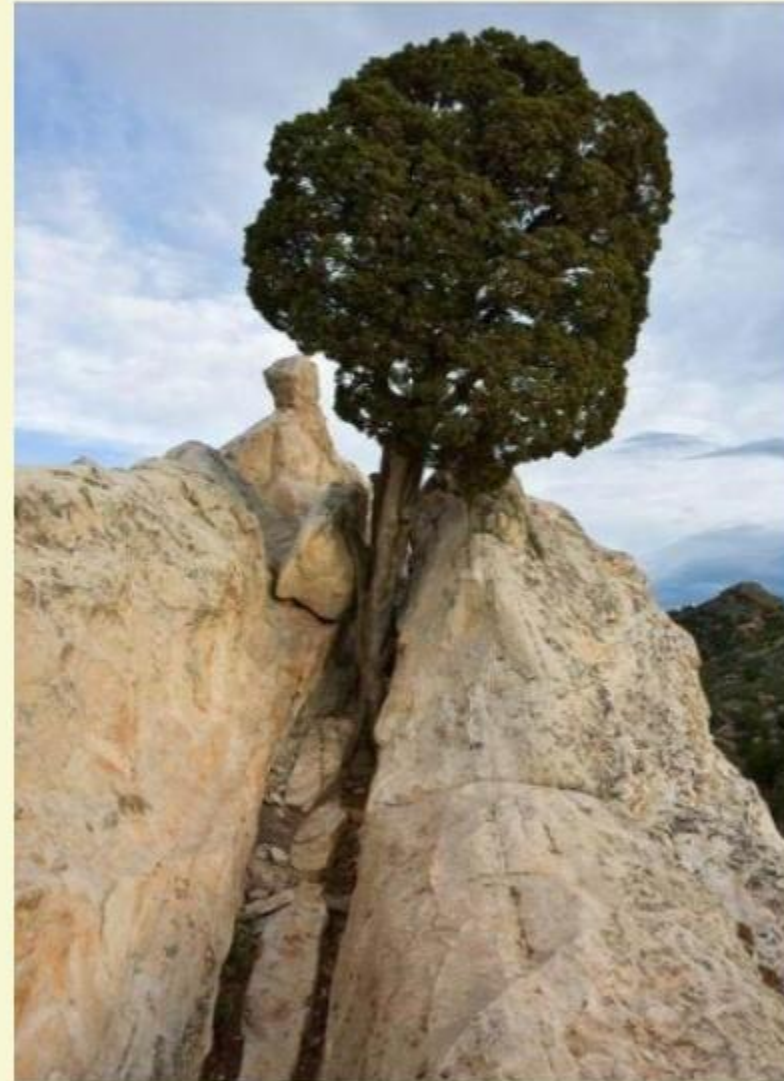
5 Algae utilize mineral nutrients for growth and help in concentration of iron and manganese oxides.

6 Plant roots exert a tremendous pressure on Earth materials mechanically breaking them apart

WEATHERING

3

BIOLOGICAL WEATHERING PROCESSES



WEATHERING

SIGNIFICANCE OF WEATHERING



1 Weathering processes are responsible for **breaking down the rocks** into smaller fragments and preparing the way for formation of **not only regolith and soils, but also erosion and mass movements.**

2 **Biomes and biodiversity** is basically a result of forests (vegetation) and forests depend upon the depth of weathering mantles.

WEATHERING

SIGNIFICANCE OF WEATHERING

WHAT DO YOU UNDERSTAND BY ENRICHMENT?

When rocks undergo weathering, some materials are removed through chemical or physical leaching by groundwater and thereby the concentration of remaining (valuable) materials increases. Without such a weathering taking place, the concentration of the same valuable material may not be sufficient and economically viable to exploit, process and refine. This is what is called enrichment

3 Erosion cannot be significant if the rocks are not weathered. That means, weathering aids mass wasting, erosion and reduction relief and changes in landforms area are a consequence of erosion.

4 Weathering of rocks and deposits helps in the enrichment and concentrations of certain valuable ores of iron, manganese, aluminum, copper etc, which are of great importance for the national economy. Weathering is an important process in the formation of soils.

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For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

OUR TEAM

G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binukrishna

G. Ravi Babu

D. Joji

K. Victor Babu

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